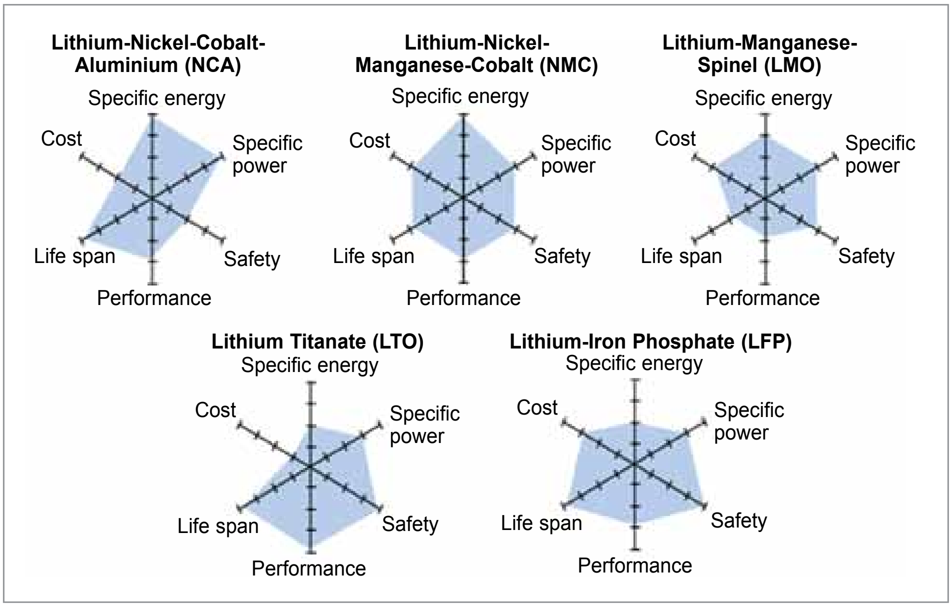

Comparisons of Li-ion battery chemistry performance parameters (Source: https://us.v-cdn.net)

each type of cell, along with their properties.

Like many other commonly available cells, Li-ion cells are made up of four components—cathode, anode, electrolyte, and separator. Li-ion cells commonly use graphite at anode and an intercalated lithium compound at cathode. The capacity and voltage of a Li-ion cell is dependent on the cathode while the charge and discharge rate of a battery depends on the anode. The amount of lithium and the active material in the cathode dictates the battery capacity and voltage while the potential difference between the anode and the cathode determines the voltage of a battery.

The active compounds in Li-ion batteries interact in complex and complementary ways. Therefore, understanding Li-ion cells' chemistry is essential for selecting the most suitable one for an application. The six most common Li-ion battery cells are described below.

LCO: Low energy density, high-risk cell

Lithium cobalt oxide (LCO), also known as lithium cobaltate, is a type of Li-ion cell whose cathode is made of lithium, cobalt, and oxygen. LCO cells have a lower energy density of 150 to 180Wh/kg and a nominal voltage of 3.7V. LCO's main element, cobalt, is an energy-dense material that is also very volatile. Hence, LCO batteries have high specific energy but are not suitable for high charge and discharge rates. Furthermore, these batteries have poor thermal stability, which makes them unsuitable for operation in harsh environments. LCO batteries are sensitive to overcharging and operation in high-temperature environments and are prone to thermal runaway.

Application. LCO batteries are more suitable for use as energy batteries than power batteries and hence are used in such applications as smartphones, cameras, etc.

LMO: Cell with low life and high C-rates

Lithium manganese oxide (LMO) batteries, or lithium-ion manganese batteries, are based on manganese oxide, which is a non-toxic and earth-abundant element, making these batteries economical. These batteries have a nominal voltage of 3.9V and an energy density of 100 to 150Wh/kg. These batteries also have an exceptionally low self-discharge rate. The lower internal resistance and high-temperature stability make these batteries a safer option for power-intensive applications. Moreover, the LMO batteries can achieve high charge and discharge rates due to their lower internal resistance. But LMO cells have a lower energy capacity and a low operational life of 300 to 700 cycles, which has resulted in limited research and advancement of this battery chemistry.

Application. LMO cells are used as power batteries in applications requiring high continuous power and are thus ideal for use in medical, military, and industrial applications. These can also be used in power tools, external defibrillators, etc.

LFP: Long lasting, affordable, and safe cell

Lithium iron phosphate cells, also known as lithium ferro-phosphate (LFP) cells, are one of the most common cell chemistries in today's time. These cells have a nominal voltage of 3.2V and an energy density of 90 to 160Wh/kg. Although having a lower energy density than many other chemistries, LFP has a very high operational life of 7,000 to 12,000 charge cycles. The LFP cells have high thermal stability, low self-discharge rates, and low internal resistance, which make LFP cells suitable for power operations that can achieve high charge and discharge rates. LFP cells are one of the most affordable chemistries, considering their long operational life.

Application. LFP cells are suitable for high-power applications and can be used in harsh conditions, making them a good option for ap-